MAULANA AZAD NATIONAL INSTITUTEOF TECHN

BHOPAL

NETWORK LAB

EXPERIMENT NO.08

Aim: To study the Reciprocity Theorem Verification.

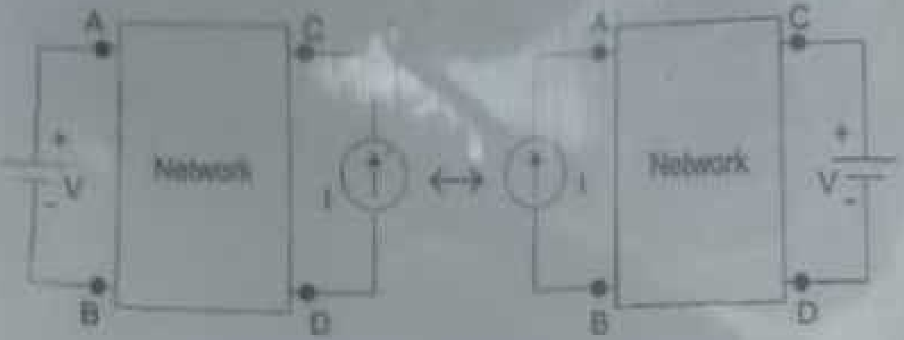
Apparatus Required:

1. Trainer kit
2. Connecting cords

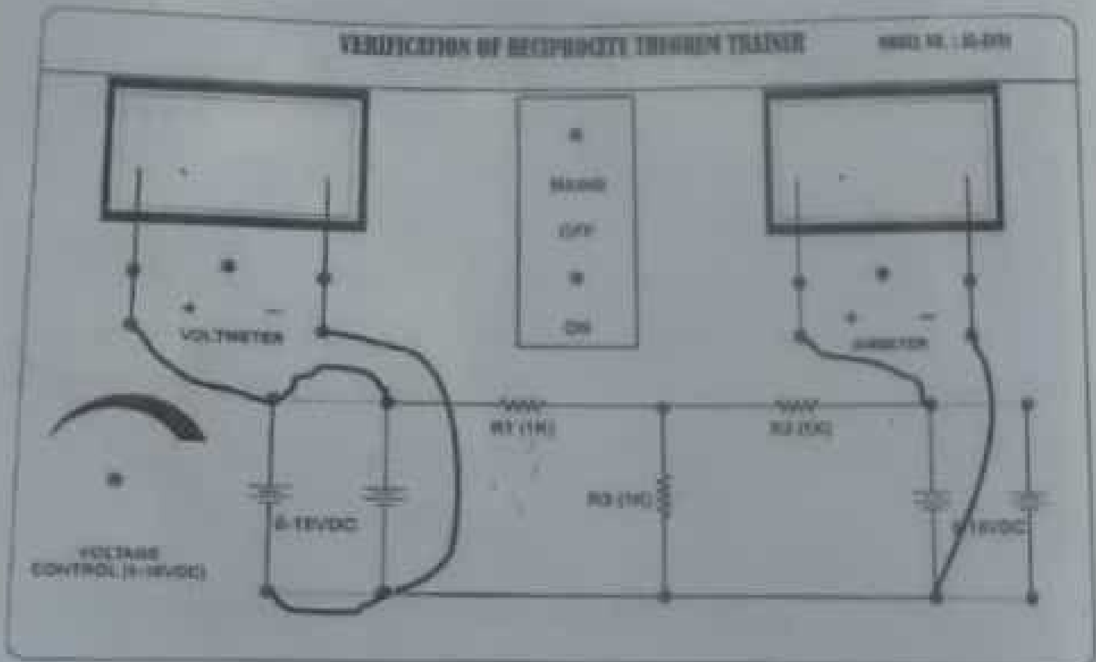
Theory:

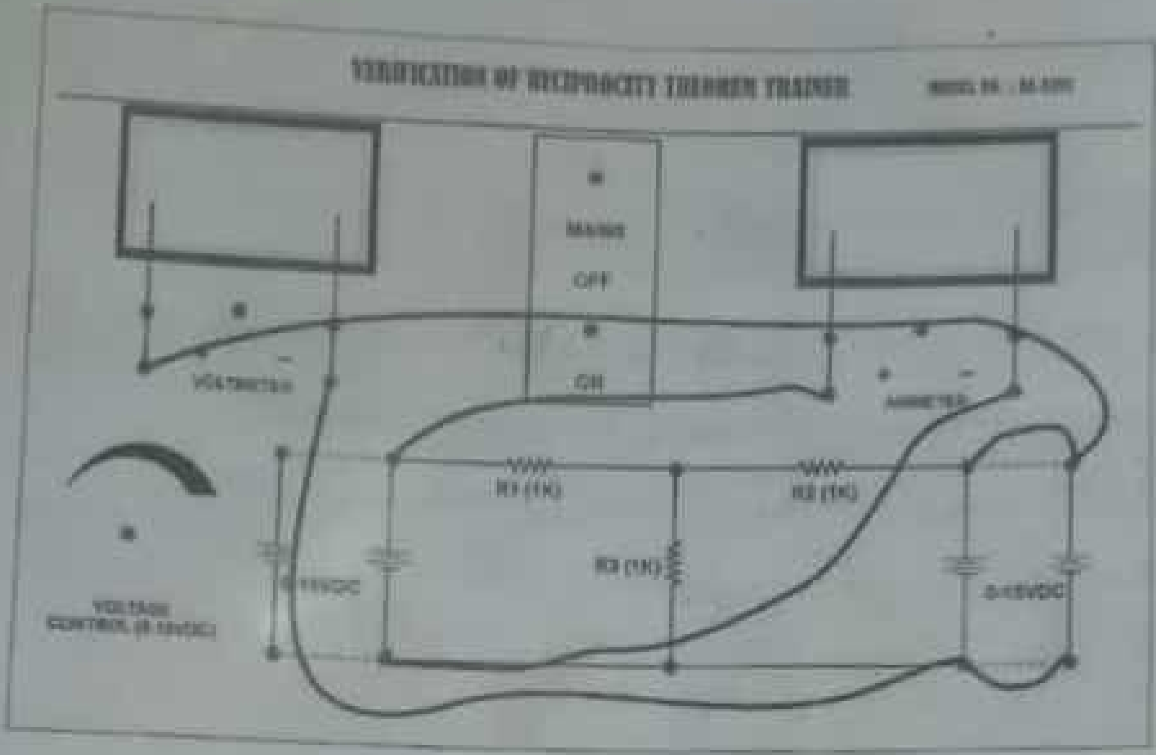
In many electrical networks it is found that if the positions of voltage source and ammeter are interchanged, the reading of ammeter remains the same. It is not clear to you. Let's explain it in details. Suppose a voltage source is connected to a passive network and an ammeter is connected to other part of the network to indicate the response. Now any one interchanges the positions of ammeter and voltage source that means he or she connects the voltage source at the part of the network where the ammeter was connected and connects ammeter to that part of the network where the voltage source was connected. The response of the ammeter means current through the ammeter would be the same in both the cases. This is where the property of reciprocity comes in the circuit. The particular circuit that has this reciprocal property, is called reciprocal circuit. This type of circuit perfectly obeys **reciprocity theorem**.

The voltage source and the ammeter used in this theorem must be ideal. That means the internal resistance of both the voltage source and ammeter must be zero. The reciprocal circuit may be a simple or complex network. But every complex reciprocal passive network can be simplified into a simple network. As per reciprocity theorem, in a linear passive network, supply voltage V and output current I are mutually transferable. The ratio of V and I is called the transfer resistance. The theorem can easily be understood by this following example.



Connection Diagram





Procedure:

1. Connect the circuit as shown in connection diagram.
2. Give dc supply in the one end and check current in the other end.
3. Then interchange the supply and check current in the end where supply has given earlier.

Conclusion:

When you interchange the source then current in the other end is same for both cases. And we said that reciprocity theorem is verified.

Questions :

1. What is Reciprocity Theorem?

2. Whether we can apply the Reciprocity theorem directly to the circuits which have multiple independent sources?

3. Whether we can apply the Reciprocity theorem to the AC circuit?

4. What are the limitations of the Reciprocity theorem?

5.Whether Antennas will satisfy the Reciprocity theorem?

6. What is a Reciprocal network?

Ans.

1. According to the reciprocity theorem, for linear and bilateral networks or circuits with only one independent source, the response to the excitation ratio is constant even if the source is swapped from the input port to the output port.
2. No, we can apply the Reciprocity theorem directly to the circuits which contain only one independent source. So, if the circuits contain multiple independent sources, then first we can apply the Superposition theorem & then Reciprocity Theorem.
3. Yes. We can apply the Reciprocity theorem to the AC circuit just like the way we applied the Reciprocity theorem to the DC circuit. If the AC circuit consists of linear and bilateral elements and only one independent source, then we can apply the Reciprocity theorem.
4. We can't apply the Reciprocity theorem if the network is having non-linear and/or unilateral elements. Similarly, we can't apply the Reciprocity theorem if the network has a dependent source because the dependent source makes the network active.
5. Yes. Antennas will satisfy the Reciprocity theorem. Hence, we can use the same antenna as either a transmitting antenna or receiving antenna based on the requirement. For effective communication, there should be impedance matching between the transmitter and receiver
6. If any electrical network satisfies the Reciprocity theorem, it is known as a Reciprocal network. All the electrical networks which contain passive elements are always Reciprocal networks because they satisfy the Reciprocity theorem.